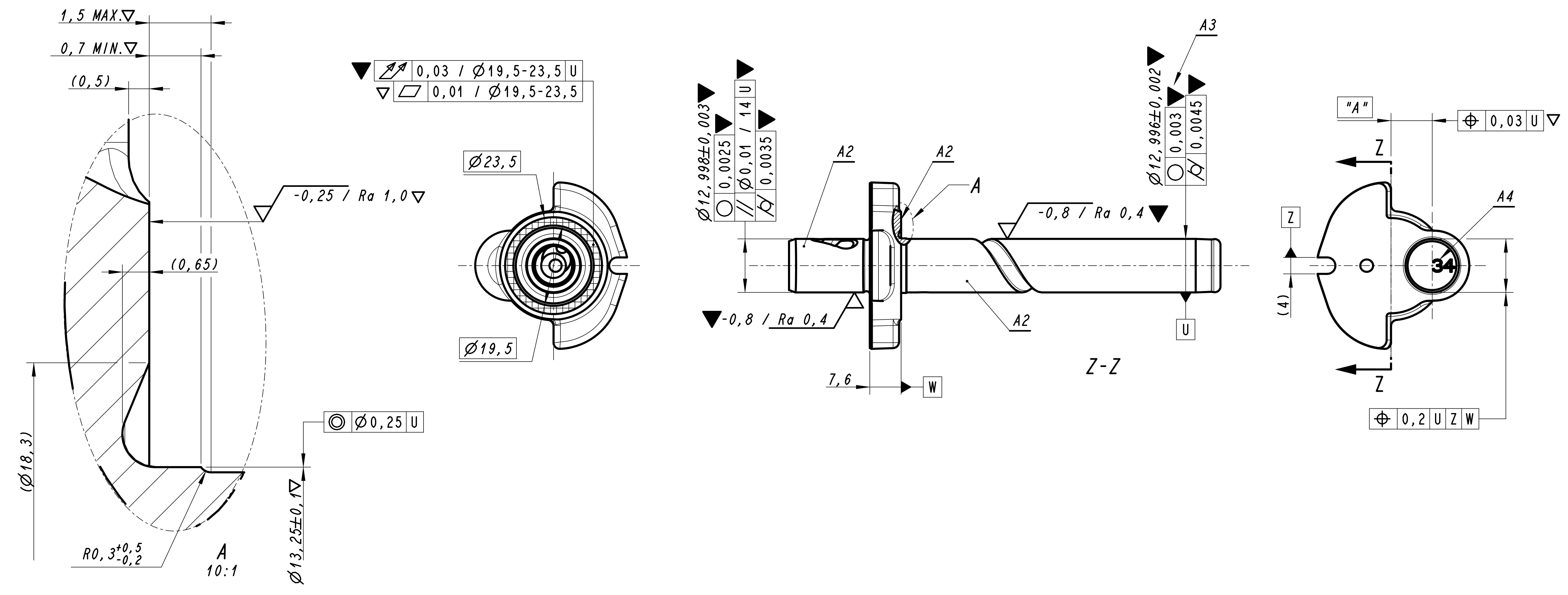
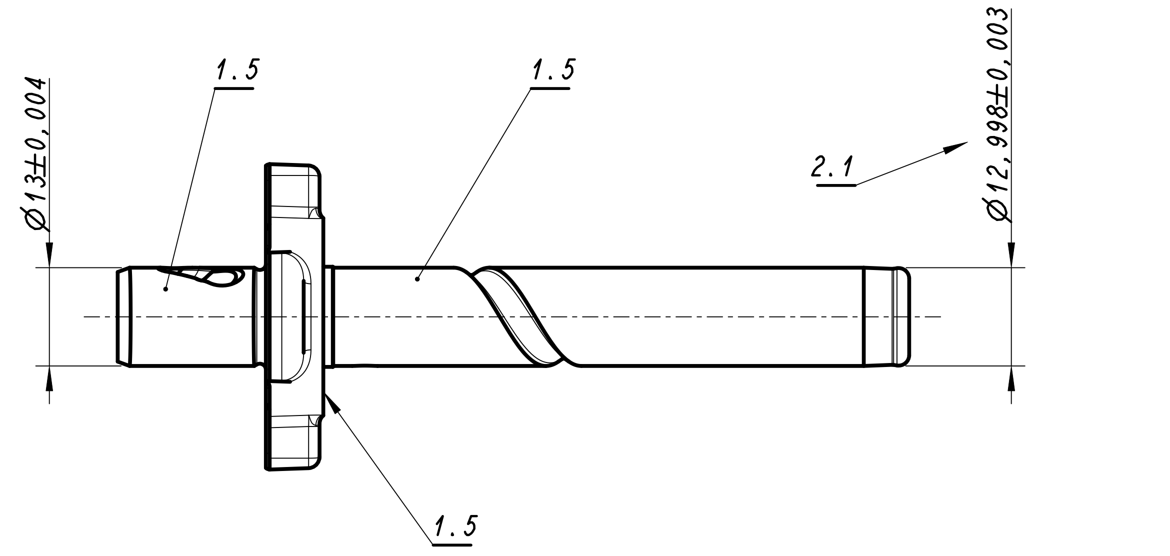
1 - GRINDING PROCESS



NOTES:
A1 - GEOMETRIC TOLERANCES ACCORDING TO TSS 001602.
A2 - REGIONS OF BEARING SURFACES MUST BE FREE OF SHOCK MARKS. ALL BORDERS, HOLES AND LUBRIFICATION GROOVES MUST HAVE SMOOTH EDGES AND FREE OF BURRS.
A3 - ROUNDNESS PROFILE CONTROL ACCORDING TO TSS 002621. ▽
A4 - IDENTIFICATION ALLOWED IN THIS AREA.

2 - PHOSPHATE PROCESS

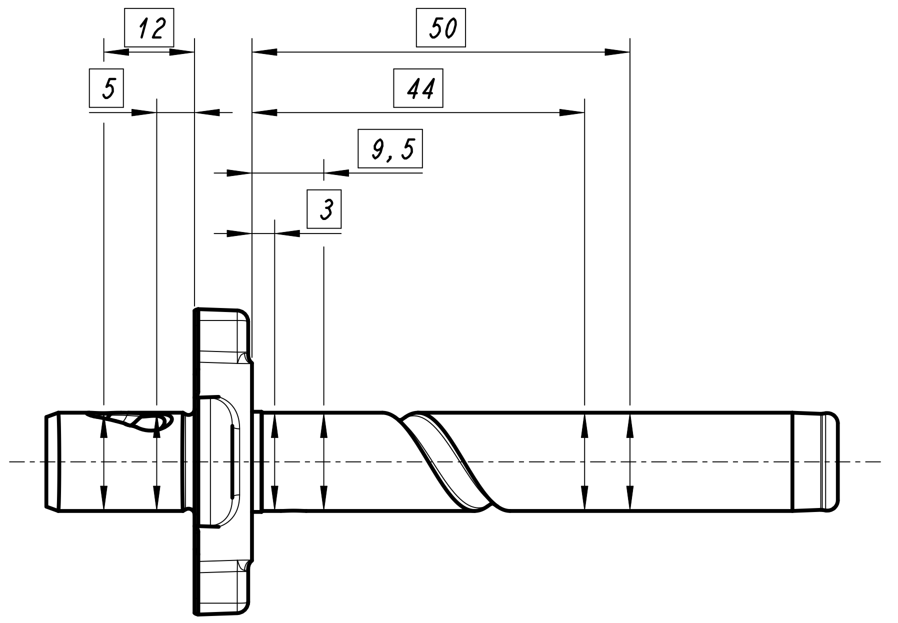


NOTES:
1 - GENERAL REQUIREMENTS:
1.1 - DRAWING ACCORDING TO ISO STANDARDS, SEE TSS 002513.
1.2 - DIMENSIONS AND SPECIFICATIONS SIGNED WITH ▽▼⊕ , SEE TSS 002470.
1.3 - REQUIREMENTS ABOUT HAZARDOUS SUBSTANCES ACCORDING TO TSS 002420.⊕
1.4 - HFC MANDATORY REQUIREMENTS ACCORDING TO TSS 002369.▼
1.5 - REGIONS OF BEARING SURFACES MUST BE FREE OF SHOCK MARKS.
1.6 - ALL BORDERS, HOLES AND LUBRIFICATION GROOVES MUST HAVE SMOOTH EDGES AND FREE OF BURRS.
1.7 - ANY INCONSISTENCY WITH THE TSS, THE DIMENSIONS OF THE DRAWING SHOULD BE CONSIDERED.

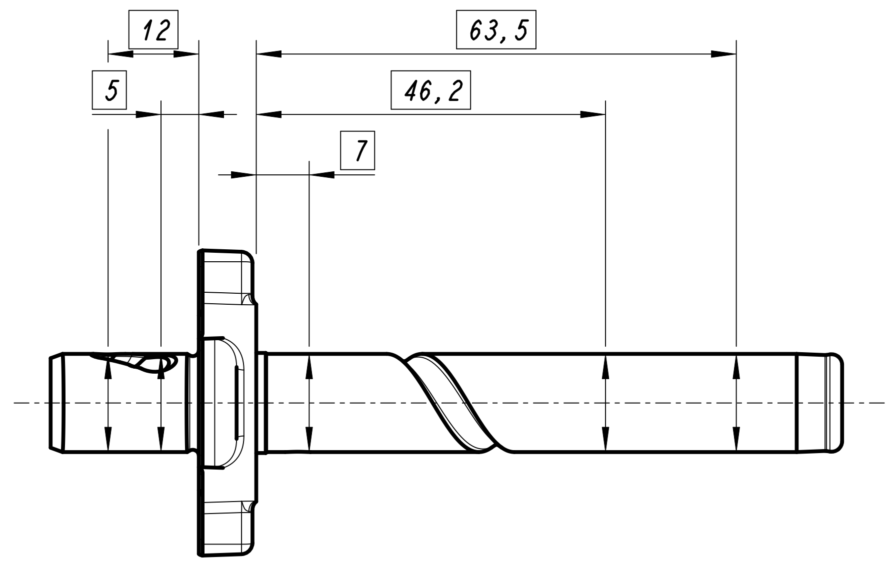
2 - GEOMETRY AND DIMENSIONAL CONTROL:
2.1 - ASSEMBLY REQUIREMENTS ACCORDING TO TSS 001559.▼

3 - MATERIAL AND SURFACE:
3.1 - SURFACE COATING: MANGANIC PHOSPHATIZING ACCORDING TO ISO 9717. COAT THICKNESS: 0,0005 TO 0,0015.

SECTIONS FOR MEASURING THE ROUNDNESS AFTER GRINDING:



SECTIONS FOR MEASURING THE DIAMETERS AFTER GRINDING AND PHOSPHATE:



CODE	"A"	MATERIAL
113342246	9,1	113392246
113344512	11,5	113394512
113344511	8,8	113394511
113344510	8,3	113394510
113344509	8,0	113394509
113344508	7,8	113394508
113344506	10,5	113394506
113344505	8,5	113394505
113344504	7,5	113394504
113344503	7,0	113394503
113344502	6,5	113394502
113344501	5,3	113394501
113342192	10,0	113392192

ECM		REVISION	LO	BY	DATE
CR26574	4	UPDATED THE DRAWING TO THE NEW LAYOUT FOR MACH. CRANKSHAFTS	-	JVB	
		INCL. DIM. (0,5)	F9		
		INCL. DIM. (Ø18,3)	E9		
		GD&T 0,03 / Ø19,5-23,5 U WAS 0,03 H	F8		
		GD&T 0,01 / Ø19,5-23,5 WAS 0,01	F8		
		INCL. DATUM REF. W	E6		
		DATUM REF. U WAS H	E5		
		INCL. DATUM REF. Z	F4		
		INCL. DIM. (4)	E4		
		DIM. "A" WAS "A"±0,015	F4		
		REMOV. HIC SYMBOL	F4		
		INCL. GD&T ⊕ 0,03 U	F4		
		INCL. HIC	F3		
		INCL. GD&T ⊕ 0,2 U Z W	E4		
		INCL. DIM. Ø13±0,004	F3		
		INCL. DIM. Ø12,998±0,003	F1		
		REMOV. DATUM REF. G	E5		
		DIM. Ø12,998±0,003 WAS Ø13±0,004	F7		
		DIM. Ø12,996±0,002 WAS Ø13 +0,001 -0,005	F5		
		INCL. NOTE 1.7	-		
		GD&T ◎ Ø0,25 U WAS ◎ Ø0,25 H	E8		
CR33117	5	INCL. 113342246	TAB	LYF	2026.06.01

MATERIAL : SEE TABLE	embraco Nidec	GEN. TOL. : ±0,2
		ANG. TOL. : ---
		SCALE : 1:1
		UNIT: mm A2
CODE :	TITLE, DOCUMENT TYPE	REPLACE: 113342192 - REV.3
DRAWN : IGOR KASPER DEDECO	CRANKSHAFT	No. 113342192
APP.: ADILSON MANKE		
ECM : CR20410 DATE : 2022.SEP.19	THIS MATERIAL IS CONFIDENTIAL AND PROPRIETARY TO EMBRACO, AND MAY NOT BE REPRODUCED, DISCLOSED, DISTRIBUTED OR USED WITHOUT PERMISSION	SHEET NUMBER: 1 OF 1